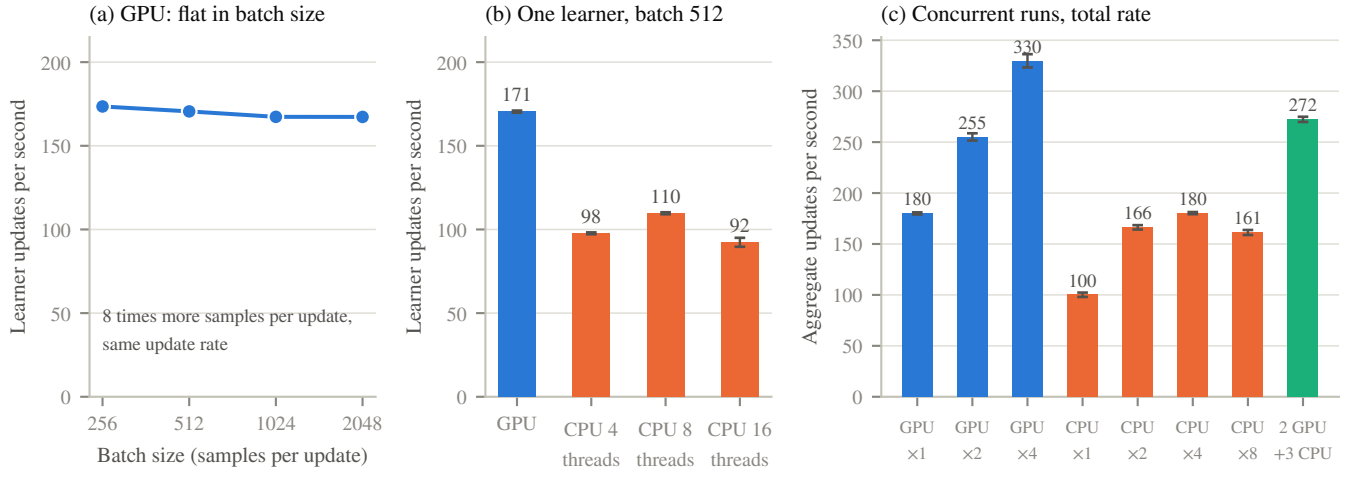
(a) GPU: flat in batch size
Learner updates per second
200
150
100
50
0
8 times more samples per update,
same update rate
256
512
1024
2048
Batch size (samples per update)
(b) One learner, batch 512
Learner updates per second
200
150
100
50
0
171
98
110
92
GPU
CPU 4 threads
CPU 8 threads
CPU 16 threads
(c) Concurrent runs, total rate
Aggregate updates per second
350
300
250
200
150
100
50
0
180
255
330
100
166
180
161
272
GPU ×1
GPU ×2
GPU ×4
CPU ×1
CPU ×2
CPU ×4
CPU ×8
2 GPU +3 CPU